

A proof of the conjectured recurrence for OEIS A001724

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Abstract

OEIS A001724 carries a conjectured linear recurrence with polynomial coefficients, of order 5. We prove it. The entry's exponential generating function is transcendental, so the usual algebraic arguments do not apply; instead we observe that it lies in a finitely generated module over $\mathbb{Q}(x)$ spanned by monomials $\log(u)^a \exp(g)^k$, and that this module is closed under differentiation, because $(\log u)' = u'/u$ and $(\exp g)' = g' \exp g$ lower the log power and fix the exponential respectively. Re-indexing the recurrence forward turns shifts into derivatives, so the sequence $b(m) = \sum_j q_j(m) a(m+j)$ has exponential generating function $B = \sum_j q_j(\theta) A^{(j)}$, computed exactly inside that module. Here B collapses to the polynomial 0, of degree 0, which proves the recurrence for every $n > 0$.

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1 The sequence and the conjecture

OEIS A001724 is “Generalized Stirling numbers, $[n+9,9]_5$ ”. It has offset 0 and begins

$$a(0), \dots, a(7) = 1, 35, 835, 17360, 342769, 6687009, 131590430, 2642422750, \dots$$

The entry gives the exponential generating function

$$A(x) = \sum_{n \geq 0} a(n) \frac{x^n}{n!} = \frac{-420 \log(1-x)^4 + 1066 \log(1-x)^3 - 753 \log(1-x)^2 + 156 \log(1-x) - 6}{6(x-1)^9}, \quad (1)$$

and it carries the following comment:

Conjecture D-finite with recurrence $a(n) + 5*(-n-6)*a(n-1) + 5*(2*n^2+22*n+61)*a(n-2) - 5*(n+5)*(2*n^2+20*n+51)*a(n-3) + (5*n^4+90*n^3+610*n^2+1845*n+2101)*a(n-4) - (n+4)^5*a(n-5)=0$.

As of the “Last modified” line on the live entry (2026-07-20, revision 43) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

Writing the conjectured relation in the form $\sum_{i=0}^5 p_i(n) a(n-i) = 0$, the coefficient polynomials are

$$p_0(n) = 1, \quad p_1(n) = -5(n+6), \quad p_2(n) = 5(2n^2 + 22n + 61), \quad p_3(n) = -5(n+5)(2n^2 + 20n + 51),$$

2 Recurrences as differential operators

Let $A(x) = \sum_{n \geq 0} a(n) x^n / n!$ be the exponential generating function of a sequence, and let $\theta = x \frac{d}{dx}$, so that θ acts on x^m as multiplication by m . For a polynomial p we write $p(\theta)$ for the corresponding differential operator, so that $p(\theta)A = \sum_m p(m) a(m) x^m / m!$.

Lemma 1. *Let $p_0, \dots, p_r$ be polynomials and suppose the conjecture asserts $\sum_{i=0}^r p_i(n) a(n-i) = 0$. Re-index by $n = m + r$ and $j = r - i$, and write $q_j(m) = p_{r-j}(m + r)$, so the claim reads $\sum_{j=0}^r q_j(m) a(m + j) = 0$. Put $b(m) = \sum_j q_j(m) a(m + j)$. Then*

$$B(x) := \sum_{m \geq 0} b(m) \frac{x^m}{m!} = \sum_{j=0}^r (q_j(\theta) A^{(j)})(x),$$

where $A^{(j)}$ denotes the j -th derivative of A .

Proof. For an exponential generating function an upward shift is differentiation: $\sum_{m \geq 0} a(m + j) x^m / m! = A^{(j)}(x)$. Multiplying the coefficient of $x^m / m!$ by $q_j(m)$ is exactly the action of $q_j(\theta)$, and summing over j gives the statement. The re-indexing is what makes this work: a downward shift $a(n - i)$ would correspond to repeated integration, whereas an upward shift is a derivative, and the field used below is closed under differentiation. $\square$

Corollary 2. *With the notation of Lemma 1, the recurrence $\sum_i p_i(n) a(n - i) = 0$ holds for every $n > d + r$ if and only if B is a polynomial of degree at most d .*

Proof. $b(m)$ is $m!$ times the coefficient of x^m in B , so $b(m) = 0$ for all $m > d$ says exactly that B has no terms above x^d . $\square$

This turns the conjecture into a closed-form identity. The point is that it can then be decided exactly, with no truncation: A is algebraic, so B lies in the same field as A , and one only has to recognise it.

Lemma 3. *Let $u, g \in \mathbb{Q}(x)$ and let M be the $\mathbb{Q}(x)$ -module spanned by the monomials $\log(u)^a \exp(g)^k$ for $a \geq 0$ and $k \in \mathbb{Z}$. Then M is closed under d/dx , and hence under $\theta = x d/dx$; explicitly,*

$$\frac{d}{dx} \left(c \log(u)^a \exp(g)^k \right) = \left(c' + c k g' \right) \log(u)^a \exp(g)^k + c a \frac{u'}{u} \log(u)^{a-1} \exp(g)^k, \quad c \in \mathbb{Q}(x).$$

Proof. Differentiate the product. The factor $\exp(g)^k = \exp(kg)$ contributes kg' and does not change the monomial; the factor $\log(u)^a$ contributes $a(u'/u) \log(u)^{a-1}$, lowering the log exponent by one. Both new coefficients lie in $\mathbb{Q}(x)$, so the result is again a finite combination of the same monomials. $\square$

Two features make this usable. The exponential part never mixes monomials at all, and the logarithmic part only ever moves *downwards*, so the module generated by any single element is finite dimensional and every computation terminates. Writing B in this basis, the test of Corollary 2 becomes: B is a polynomial precisely when every coefficient outside the monomial $\log(u)^0 \exp(g)^0$ vanishes and the remaining coefficient has constant denominator. Both are decided by exact cancellation of rational functions, so no series truncation enters anywhere.

3 The computation for A001724

The generating function (1) lies in the module M of Lemma 3. Applying Lemma 1 to the coefficient polynomials of Section 1 and reducing in K by Lemma 3, the $\sqrt{D}$ component of B cancels identically and the rational part collapses to

$$B(x) = 0,$$

a polynomial of degree 0.

Theorem 4. *The conjectured recurrence*

$1 a(n)-5 (n+6) a(n-1)+5 (2n^2+22n+61) a(n-2)-5 (n+5) (2n^2+20n+51) a(n-3)+(5n^4+90n^3+610n^2-115n-120) a(n-4)+5 (n+4) (2n^2+18n+47) a(n-5)-5 (n+3) (2n^2+16n+43) a(n-6)+5 (n+2) (2n^2+14n+39) a(n-7)-5 (n+1) (2n^2+12n+35) a(n-8)+5 n (2n^2+10n+31) a(n-9)-5 n (2n^2+8n+27) a(n-10)+5 n (2n^2+6n+23) a(n-11)-5 n (2n^2+4n+19) a(n-12)+5 n (2n^2+2n+15) a(n-13)+5 n (2n^2) a(n-14)+5 n (2n^2-2n+1) a(n-15)+5 n (2n^2-4n+3) a(n-16)+5 n (2n^2-6n+5) a(n-17)+5 n (2n^2-8n+7) a(n-18)+5 n (2n^2-10n+9) a(n-19)+5 n (2n^2-12n+11) a(n-20)+5 n (2n^2-14n+13) a(n-21)+5 n (2n^2-16n+15) a(n-22)+5 n (2n^2-18n+17) a(n-23)+5 n (2n^2-20n+19) a(n-24)+5 n (2n^2-22n+21) a(n-25)+5 n (2n^2-24n+23) a(n-26)+5 n (2n^2-26n+25) a(n-27)+5 n (2n^2-28n+27) a(n-28)+5 n (2n^2-30n+29) a(n-29)+5 n (2n^2-32n+31) a(n-30)+5 n (2n^2-34n+33) a(n-31)+5 n (2n^2-36n+35) a(n-32)+5 n (2n^2-38n+37) a(n-33)+5 n (2n^2-40n+39) a(n-34)+5 n (2n^2-42n+41) a(n-35)+5 n (2n^2-44n+43) a(n-36)+5 n (2n^2-46n+45) a(n-37)+5 n (2n^2-48n+47) a(n-38)+5 n (2n^2-50n+49) a(n-39)+5 n (2n^2-52n+51) a(n-40)+5 n (2n^2-54n+53) a(n-41)+5 n (2n^2-56n+55) a(n-42)+5 n (2n^2-58n+57) a(n-43)+5 n (2n^2-60n+59) a(n-44)+5 n (2n^2-62n+61) a(n-45)+5 n (2n^2-64n+63) a(n-46)+5 n (2n^2-66n+65) a(n-47)+5 n (2n^2-68n+67) a(n-48)+5 n (2n^2-70n+69) a(n-49)+5 n (2n^2-72n+71) a(n-50)+5 n (2n^2-74n+73) a(n-51)+5 n (2n^2-76n+75) a(n-52)+5 n (2n^2-78n+77) a(n-53)+5 n (2n^2-80n+79) a(n-54)+5 n (2n^2-82n+81) a(n-55)+5 n (2n^2-84n+83) a(n-56)+5 n (2n^2-86n+85) a(n-57)+5 n (2n^2-88n+87) a(n-58)+5 n (2n^2-90n+89) a(n-59)+5 n (2n^2-92n+91) a(n-60)+5 n (2n^2-94n+93) a(n-61)+5 n (2n^2-96n+95) a(n-62)+5 n (2n^2-98n+97) a(n-63)+5 n (2n^2-100n+99) a(n-64)+5 n (2n^2-102n+101) a(n-65)+5 n (2n^2-104n+103) a(n-66)+5 n (2n^2-106n+105) a(n-67)+5 n (2n^2-108n+107) a(n-68)+5 n (2n^2-110n+109) a(n-69)+5 n (2n^2-112n+111) a(n-70)+5 n (2n^2-114n+113) a(n-71)+5 n (2n^2-116n+115) a(n-72)+5 n (2n^2-118n+117) a(n-73)+5 n (2n^2-120n+119) a(n-74)+5 n (2n^2-122n+121) a(n-75)+5 n (2n^2-124n+123) a(n-76)+5 n (2n^2-126n+125) a(n-77)+5 n (2n^2-128n+127) a(n-78)+5 n (2n^2-130n+129) a(n-79)+5 n (2n^2-132n+131) a(n-80)+5 n (2n^2-134n+133) a(n-81)+5 n (2n^2-136n+135) a(n-82)+5 n (2n^2-138n+137) a(n-83)+5 n (2n^2-140n+139) a(n-84)+5 n (2n^2-142n+141) a(n-85)+5 n (2n^2-144n+143) a(n-86)+5 n (2n^2-146n+145) a(n-87)+5 n (2n^2-148n+147) a(n-88)+5 n (2n^2-150n+149) a(n-89)+5 n (2n^2-152n+151) a(n-90)+5 n (2n^2-154n+153) a(n-91)+5 n (2n^2-156n+155) a(n-92)+5 n (2n^2-158n+157) a(n-93)+5 n (2n^2-160n+159) a(n-94)+5 n (2n^2-162n+161) a(n-95)+5 n (2n^2-164n+163) a(n-96)+5 n (2n^2-166n+165) a(n-97)+5 n (2n^2-168n+167) a(n-98)+5 n (2n^2-170n+169) a(n-99)+5 n (2n^2-172n+171) a(n-100)+5 n (2n^2-174n+173) a(n-101)+5 n (2n^2-176n+175) a(n-102)+5 n (2n^2-178n+177) a(n-103)+5 n (2n^2-180n+179) a(n-104)+5 n (2n^2-182n+181) a(n-105)+5 n (2n^2-184n+183) a(n-106)+5 n (2n^2-186n+185) a(n-107)+5 n (2n^2-188n+187) a(n-108)+5 n (2n^2-190n+189) a(n-109)+5 n (2n^2-192n+191) a(n-110)+5 n (2n^2-194n+193) a(n-111)+5 n (2n^2-196n+195) a(n-112)+5 n (2n^2-198n+197) a(n-113)+5 n (2n^2-200n+199) a(n-114)+5 n (2n^2-202n+201) a(n-115)+5 n (2n^2-204n+203) a(n-116)+5 n (2n^2-206n+205) a(n-117)+5 n (2n^2-208n+207) a(n-118)+5 n (2n^2-210n+209) a(n-119)+5 n (2n^2-212n+211) a(n-120)+5 n (2n^2-214n+213) a(n-121)+5 n (2n^2-216n+215) a(n-122)+5 n (2n^2-218n+217) a(n-123)+5 n (2n^2-220n+219) a(n-124)+5 n (2n^2-222n+221) a(n-125)+5 n (2n^2-224n+223) a(n-126)+5 n (2n^2-226n+225) a(n-127)+5 n (2n^2-228n+227) a(n-128)+5 n (2n^2-230n+229) a(n-129)+5 n (2n^2-232n+231) a(n-130)+5 n (2n^2-234n+233) a(n-131)+5 n (2n^2-236n+235) a(n-132)+5 n (2n^2-238n+237) a(n-133)+5 n (2n^2-240n+239) a(n-134)+5 n (2n^2-242n+241) a(n-135)+5 n (2n^2-244n+243) a(n-136)+5 n (2n^2-246n+245) a(n-137)+5 n (2n^2-248n+247) a(n-138)+5 n (2n^2-250n+249) a(n-139)+5 n (2n^2-252n+251) a(n-140)+5 n (2n^2-254n+253) a(n-141)+5 n (2n^2-256n+255) a(n-142)+5 n (2n^2-258n+257) a(n-143)+5 n (2n^2-260n+259) a(n-144)+5 n (2n^2-262n+261) a(n-145)+5 n (2n^2-264n+263) a(n-146)+5 n (2n^2-266n+265) a(n-147)+5 n (2n^2-268n+267) a(n-148)+5 n (2n^2-270n+269) a(n-149)+5 n (2n^2-272n+271) a(n-150)+5 n (2n^2-274n+273) a(n-151)+5 n (2n^2-276n+275) a(n-152)+5 n (2n^2-278n+277) a(n-153)+5 n (2n^2-280n+279) a(n-154)+5 n (2n^2-282n+281) a(n-155)+5 n (2n^2-284n+283) a(n-156)+5 n (2n^2-286n+285) a(n-157)+5 n (2n^2-288n+287) a(n-158)+5 n (2n^2-290n+289) a(n-159)+5 n (2n^2-292n+291) a(n-160)+5 n (2n^2-294n+293) a(n-161)+5 n (2n^2-296n+295) a(n-162)+5 n (2n^2-298n+297) a(n-163)+5 n (2n^2-300n+299) a(n-164)+5 n (2n^2-302n+301) a(n-165)+5 n (2n^2-304n+303) a(n-166)+5 n (2n^2-306n+305) a(n-167)+5 n (2n^2-308n+307) a(n-168)+5 n (2n^2-310n+309) a(n-169)+5 n (2n^2-312n+311) a(n-170)+5 n (2n^2-314n+313) a(n-171)+5 n (2n^2-316n+315) a(n-172)+5 n (2n^2-318n+317) a(n-173)+5 n (2n^2-320n+319) a(n-174)+5 n (2n^2-322n+321) a(n-175)+5 n (2n^2-324n+323) a(n-176)+5 n (2n^2-326n+325) a(n-177)+5 n (2n^2-328n+327) a(n-178)+5 n (2n^2-330n+329) a(n-179)+5 n (2n^2-332n+331) a(n-180)+5 n (2n^2-334n+333) a(n-181)+5 n (2n^2-336n+335) a(n-182)+5 n (2n^2-338n+337) a(n-183)+5 n (2n^2-340n+339) a(n-184)+5 n (2n^2-342n+341) a(n-185)+5 n (2n^2-344n+343) a(n-186)+5 n (2n^2-346n+345) a(n-187)+5 n (2n^2-348n+347) a(n-188)+5 n (2n^2-350n+349) a(n-189)+5 n (2n^2-352n+351) a(n-190)+5 n (2n^2-354n+353) a(n-191)+5 n (2n^2-356n+355) a(n-192)+5 n (2n^2-358n+357) a(n-193)+5 n (2n^2-360n+359) a(n-194)+5 n (2n^2-362n+361) a(n-195)+5 n (2n^2-364n+363) a(n-196)+5 n (2n^2-366n+365) a(n-197)+5 n (2n^2-368n+367) a(n-198)+5 n (2n^2-370n+369) a(n-199)+5 n (2n^2-372n+371) a(n-200)+5 n (2n^2-374n+373) a(n-201)+5 n (2n^2-376n+375) a(n-202)+5 n (2n^2-378n+377) a(n-203)+5 n (2n^2-380n+379) a(n-204)+5 n (2n^2-382n+381) a(n-205)+5 n (2n^2-384n+383) a(n-206)+5 n (2n^2-386n+385) a(n-207)+5 n (2n^2-388n+387) a(n-208)+5 n (2n^2-390n+389) a(n-209)+5 n (2n^2-392n+391) a(n-210)+5 n (2n^2-394n+393) a(n-211)+5 n (2n^2-396n+395) a(n-212)+5 n (2n^2-398n+397) a(n-213)+5 n (2n^2-400n+399) a(n-214)+5 n (2n^2-402n+401) a(n-215)+5 n (2n^2-404n+403) a(n-216)+5 n (2n^2-406n+405) a(n-217)+5 n (2n^2-408n+407) a(n-218)+5 n (2n^2-410n+409) a(n-219)+5 n (2n^2-412n+411) a(n-220)+5 n (2n^2-414n+413) a(n-221)+5 n (2n^2-416n+415) a(n-222)+5 n (2n^2-418n+417) a(n-223)+5 n (2n^2-420n+419) a(n-224)+5 n (2n^2-422n+421) a(n-225)+5 n (2n^2-424n+423) a(n-226)+5 n (2n^2-426n+425) a(n-227)+5 n (2n^2-428n+427) a(n-228)+5 n (2n^2-430n+429) a(n-229)+5 n (2n^2-432n+431) a(n-230)+5 n (2n^2-434n+433) a(n-231)+5 n (2n^2-436n+435) a(n-232)+5 n (2n^2-438n+437) a(n-233)+5 n (2n^2-440n+439) a(n-234)+5 n (2n^2-442n+441) a(n-235)+5 n (2n^2-444n+443) a(n-236)+5 n (2n^2-446n+445) a(n-237)+5 n (2n^2-448n+447) a(n-238)+5 n (2n^2-450n+449) a(n-239)+5 n (2n^2-452n+451) a(n-240)+5 n (2n^2-454n+453) a(n-241)+5 n (2n^2-456n+455) a(n-242)+5 n (2n^2-458n+457) a(n-243)+5 n (2n^2-460n+459) a(n-244)+5 n (2n^2-462n+461) a(n-245)+5 n (2n^2-464n+463) a(n-246)+5 n (2n^2-466n+465) a(n-247)+5 n (2n^2-468n+467) a(n-248)+5 n (2n^2-470n+469) a(n-249)+5 n (2n^2-472n+471) a(n-250)+5 n (2n^2-474n+473) a(n-251)+5 n (2n^2-476n+475) a(n-252)+5 n (2n^2-478n+477) a(n-253)+5 n (2n^2-480n+479) a(n-254)+5 n (2n^2-482n+481) a(n-255)+5 n (2n^2-484n+483) a(n-256)+5 n (2n^2-486n+485) a(n-257)+5 n (2n^2-488n+487) a(n-258)+5 n (2n^2-490n+489) a(n-259)+5 n (2n^2-492n+491) a(n-260)+5 n (2n^2-494n+493) a(n-261)+5 n (2n^2-496n+495) a(n-262)+5 n (2n^2-498n+497) a(n-263)+5 n (2n^2-500n+499) a(n-264)+5 n (2n^2-502n+501) a(n-265)+5 n (2n^2-504n+503) a(n-266)+5 n (2n^2-506n+505) a(n-267)+5 n (2n^2-508n+507) a(n-268)+5 n (2n^2-510n+509) a(n-269)+5 n (2n^2-512n+511) a(n-270)+5 n (2n^2-514n+513) a(n-271)+5 n (2n^2-516n+515) a(n-272)+5 n (2n^2-518n+517) a(n-273)+5 n (2n^2-520n+519) a(n-274)+5 n (2n^2-522n+521) a(n-275)+5 n (2n^2-524n+523) a(n-276)+5 n (2n^2-526n+525) a(n-277)+5 n (2n^2-528n+527) a(n-278)+5 n (2n^2-530n+529) a(n-279)+5 n (2n^2-532n+531) a(n-280)+5 n (2n^2-534n+533) a(n-281)+5 n (2n^2-536n+535) a(n-282)+5 n (2n^2-538n+537) a(n-283)+5 n (2n^2-540n+539) a(n-284)+5 n (2n^2-542n+541) a(n-285)+5 n (2n^2-544n+543) a(n-286)+5 n (2n^2-546n+545) a(n-287)+5 n (2n^2-548n+547) a(n-288)+5 n (2n^2-550n+549) a(n-289)+5 n (2n^2-552n+551) a(n-290)+5 n (2n^2-554n+553) a(n-291)+5 n (2n^2-556n+555) a(n-292)+5 n (2n^2-558n+557) a(n-293)+5 n (2n^2-560n+559) a(n-294)+5 n (2n^2-562n+561) a(n-295)+5 n (2n^2-564n+563) a(n-296)+5 n (2n^2-566n+565) a(n-297)+5 n (2n^2-568n+567) a(n-298)+5 n (2n^2-570n+569) a(n-299)+5 n (2n^2-572n+571) a(n-300)+5 n (2n^2-574n+573) a(n-301)+5 n (2n^2-576n+575) a(n-302)+5 n (2n^2-578n+577) a(n-303)+5 n (2n^2-580n+579) a(n-304)+5 n (2n^2-582n+581) a(n-305)+5 n (2n^2-584n+583) a(n-306)+5 n (2n^2-586n+585) a(n-307)+5 n (2n^2-588n+587) a(n-308)+5 n (2n^2-590n+589) a(n-309)+5 n (2n^2-592n+591) a(n-310)+5 n (2n^2-594n+593) a(n-311)+5 n (2n^2-596n+595) a(n-312)+5 n (2n^2-598n+597) a(n-313)+5 n (2n^2-600n+599) a(n-314)+5 n (2n^2-602n+601) a(n-315)+5 n (2n^2-604n+603) a(n-316)+5 n (2n^2-606n+605) a(n-317)+5 n (2n^2-608n+607) a(n-318)+5 n (2n^2-610n+609) a(n-319)+5 n (2n^2-612n+611) a(n-320)+5 n (2n^2-614n+613) a(n-321)+5 n (2n^2-616n+615) a(n-322)+5 n (2n^2-618n+617) a(n-323)+5 n (2n^2-620n+619) a(n-324)+5 n (2n^2-622n+621) a(n-325)+5 n (2n^2-624n+623) a(n-326)+5 n (2n^2-626n+625) a(n-327)+5 n (2n^2-628n+627) a(n-328)+5 n (2n^2-630n+629) a(n-329)+5 n (2n^2-632n+631) a(n-330)+5 n (2n^2-634n+633) a(n-331)+5 n (2n^2-636n+635) a(n-332)+5 n (2n^2-638n+637) a(n-333)+5 n (2n^2-640n+639) a(n-334)+5 n (2n^2-642n+641) a(n-335)+5 n (2n^2-644n+643) a(n-336)+5 n (2n^2-646n+645) a(n-337)+5 n (2n^2-648n+647) a(n-338)+5 n (2n^2-650n+649) a(n-339)+5 n (2n^2-652n+651) a(n-340)+5 n (2n^2-654n+653) a(n-341)+5 n (2n^2-656n+655) a(n-342)+5 n (2n^2-658n+657) a(n-343)+5 n (2n^2-660n+659) a(n-344)+5 n (2n^2-662n+661) a(n-345)+5 n (2n^2-664n+663) a(n-346)+5 n (2n^2-666n+665) a(n-347)+5 n (2n^2-668n+667) a(n-348)+5 n (2n^2-670n+669) a(n-349)+5 n (2n^2-672n+671) a(n-350)+5 n (2n^2-674n+673) a(n-351)+5 n (2n^2-676n+675) a(n-352)+5 n (2n^2-678n+677) a(n-353)+5 n (2n^2-680n+679) a(n-354)+5 n (2n^2-682n+681) a(n-355)+5 n (2n^2-684n+683) a(n-356)+5 n (2n^2-686n+685) a(n-357)+5 n (2n^2-688n+687) a(n-358)+5 n (2n^2-690n+689) a(n-359)+5 n (2n^2-692n+691) a(n-360)+5 n (2n^2-694n+693) a(n-361)+5 n (2n^2-696n+695) a(n-362)+5 n (2n^2-698n+697) a(n-363)+5 n (2n^2-700n+699) a(n-364)+5 n (2n^2-702n+701) a(n-365)+5 n (2n^2-704n+703) a(n-366)+5 n (2n^2-706n+705) a(n-367)+5 n (2n^2-708n+707) a(n-368)+5 n (2n^2-710n+709) a(n-369)+5 n (2n^2-712n+711) a(n-370)+5 n (2n^2-714n+713) a(n-371)+5 n (2n^2-716n+715) a(n-372)+5 n (2n^2-718n+717) a(n-373)+5 n (2n^2-720n+719) a(n-374)+5 n (2n^2-722n+721) a(n-375)+5 n (2n^2-724n+723) a(n-376)+5 n (2n^2-726n+725) a(n-377)+5 n (2n^2-728n+727) a(n-378)+5 n (2n^2-730n+729) a(n-379)+5 n (2n^2-732n+731) a(n-380)+5 n (2n^2-734n+733) a(n-381)+5 n (2n^2-736n+735) a(n-382)+5 n (2n^2-738n+737) a(n-383)+5 n (2n^2-740n+739) a(n-384)+5 n (2n^2-742n+741) a(n-385)+5 n (2n^2-744n+743) a(n-386)+5 n (2n^2-746n+745) a(n-387)+5 n (2n^2-748n+747) a(n-388)+5 n (2n^2-750n+749) a(n-389)+5 n (2n^2-752n+751) a(n-390)+5 n (2n^2-754n+753) a(n-391)+5 n (2n^2-756n+755) a(n-392)+5 n (2n^2-758n+757) a(n-393)+5 n (2n^2-760n+759) a(n-394)+5 n (2n^2-762n+761) a(n-395)+5 n (2n^2-764n+763) a(n-396)+5 n (2n^2-766n+765) a(n-397)+5 n (2n^2-768n+767) a(n-398)+5 n (2n^2-770n+769) a(n-399)+5 n (2n^2-772n+771) a(n-400)+5 n (2n^2-774n+773) a(n-401)+5 n (2n^2-776n+775) a(n-402)+5 n (2n^2-778n+777) a(n-403)+5 n (2n^2-780n+779) a(n-404)+5 n (2n^2-782n+781) a(n-405)+5 n (2n^2-784n+783) a(n-406)+5 n (2n^2-786n+785) a(n-407)+5 n (2n^2-788n+787) a(n-408)+5 n (2n^2-790n+789) a(n-409)+5 n (2n^2-792n+791) a(n-410)+5 n (2n^2-794n+793) a(n-411)+5 n (2n^2-796n+795) a(n-412)+5 n (2n^2-798n+797) a(n-413)+5 n (2n^2-800n+799) a(n-414)+5 n (2n^2-802n+801) a(n-415)+5 n (2n^2-804n+803) a(n-416)+5 n (2n^2-806n+805) a(n-417)+5 n (2n^2-808n+807) a(n-418)+5 n (2n^2-810n+809) a(n-419)+5 n (2n^2-812n+811) a(n-420)+5 n (2n^2-814n+813) a(n-421)+5 n (2n^2-816n+815) a(n-422)+5 n (2n^2-818n+817) a(n-423)+5 n (2n^2-820n+819) a(n-424)+5 n (2n^2-822n+821) a(n-425)+5 n (2n^2-824n+823) a(n-426)+5 n (2n^2-826n+825) a(n-427)+5 n (2n^2-828n+827) a(n-428)+5 n (2n^2-830n+829) a(n-429)+5 n (2n^2-832n+831) a(n-430)+5 n (2n^2-834n+833) a(n-431)+5 n (2n^2-836n+835) a(n-432)+5 n (2n^2-838n+837) a(n-433)+5 n (2n^2-840n+839) a(n-434)+5 n (2n^2-842n+841) a(n-435)+5 n (2n^2-844n+843) a(n-436)+5 n (2n^2-846n+845) a(n-437)+5 n (2n^2-848n+847) a(n-438)+5 n (2n^2-850n+849) a(n-439)+5 n (2n^2-852n+851) a(n-440)+5 n (2n^2-854n+853) a(n-441)+5 n (2n^2-856n+855) a(n-442)+5 n (2n^2-858n+857) a(n-443)+5 n (2n^2-860n+859) a(n-444)+5 n (2n^2-862n+861) a(n-445)+5 n (2n^2-864n+863) a(n-446)+5 n (2n^2-866n+865) a(n-447)+5 n (2n^2-868n+867) a(n-448)+5 n (2n^2-870n+869) a(n-449)+5 n (2n^2-872n+871) a(n-450)+5 n (2n^2-874n+873) a(n-451)+5 n (2n^2-876n+875) a(n-452)+5 n (2n^2-878n+877) a(n-453)+5 n (2n^2-880n+879) a(n-454)+5 n (2n^2-882n+881) a(n-455)+5 n (2n^2-884n+883) a(n-456)+5 n (2n^2-886n+885) a(n-457)+5 n (2n^2-888n+887) a(n-458)+5 n (2n^2-890n+889) a(n-459)+5 n (2n^2-892n+891) a(n-460)+5 n (2n^2-894n+893) a(n-461)+5 n (2n^2-896n+895) a(n-462)+5 n (2n^2-898n+897) a(n-463)+5 n (2n^2-900n+899) a(n-464)+5 n (2n^2-902n+901) a(n-465)+5 n (2n^2-904n+903) a(n-466)+5 n (2n^2-906n+905) a(n-467)+5 n (2n^2-908n+907) a(n-468)+5 n (2n^2-910n+909) a(n-469)+5 n (2n^2-912n+911) a(n-470)+5 n (2n^2-914n+913) a(n-471)+5 n (2n^2-916n+915) a(n-472)+5 n (2n^2-918n+917) a(n-473)+5 n (2n^2-920n+919) a(n-474)+5 n (2n^2-922n+921) a(n-475)+5 n (2n^2-924n+923) a(n-476)+5 n (2n^2-926n+925) a(n-477)+5 n (2n^2-928n+927) a(n-478)+5 n (2n^2-930n+929) a(n-479)+5 n (2n^2-932n+931) a(n-480)+5 n (2n^2-934n+933) a(n-481)+5 n (2n^2-936n+935) a(n-482)+5 n (2n^2-938n+937) a(n-483)+5 n (2n^2-940n+939) a(n-484)+5 n (2n^2-942n+941) a(n-485)+5 n (2n^2-944n+943) a(n-486)+5 n (2n^2-946n+945) a(n-487)+5 n (2n^2-948n+947) a(n-488)+5 n (2n^2-950n+949) a(n-489)+5 n (2n^2-952n+951) a(n-490)+5 n (2n^2-954n+953) a(n-491)+5 n (2n^2-956n+955) a(n-492)+5 n (2n^2-958n+957) a(n-493)+5 n (2n^2-960n+959) a(n-494)+5 n (2n^2-962n+961) a(n-495)+5 n (2n^2-964n+963) a(n-496)+5 n (2n^2-966n+965) a(n-497)+5 n (2n^2-968n+967) a(n-498)+5 n (2n^2-970n+969) a(n-499)+5 n (2n^2-972n+971) a(n-500)+5 n (2n^2-974n+973) a(n-501)+5 n (2n^2-976n+975) a(n-502)+5 n (2n^2-978n+977) a(n-503)+5 n (2n^2-980n+979) a(n-504)+5 n (2n^2-982n+981) a(n-505)+5 n (2n^2-984n+983) a(n-506)+5 n (2n^2-986n+985) a(n-507)+5 n (2n^2-988n+987) a(n-508)+5 n (2n^2-990n+989) a(n-509)+5 n (2n^2-992n+991) a(n-510)+5 n (2n^2-994n+993) a(n-511)+5 n (2n^2-996n+995) a(n-512)+5 n (2n^2-998n+997) a(n-513)+5 n (2n^2-1000n+999) a(n-514)+5 n (2n^2-1002n+1001) a(n-515)+5 n (2n^2-1004n+1003) a(n-516)+5 n (2n^2-1006n+1005) a(n-517)+5 n (2n^2-1008n+1007) a(n-518)+5 n (2n^2-1010n+1009) a(n-519)+5 n (2n^2-1012n+1011) a(n-520)+5 n (2n^2-1014n+1013) a(n-521)+5 n (2n^2-10$